

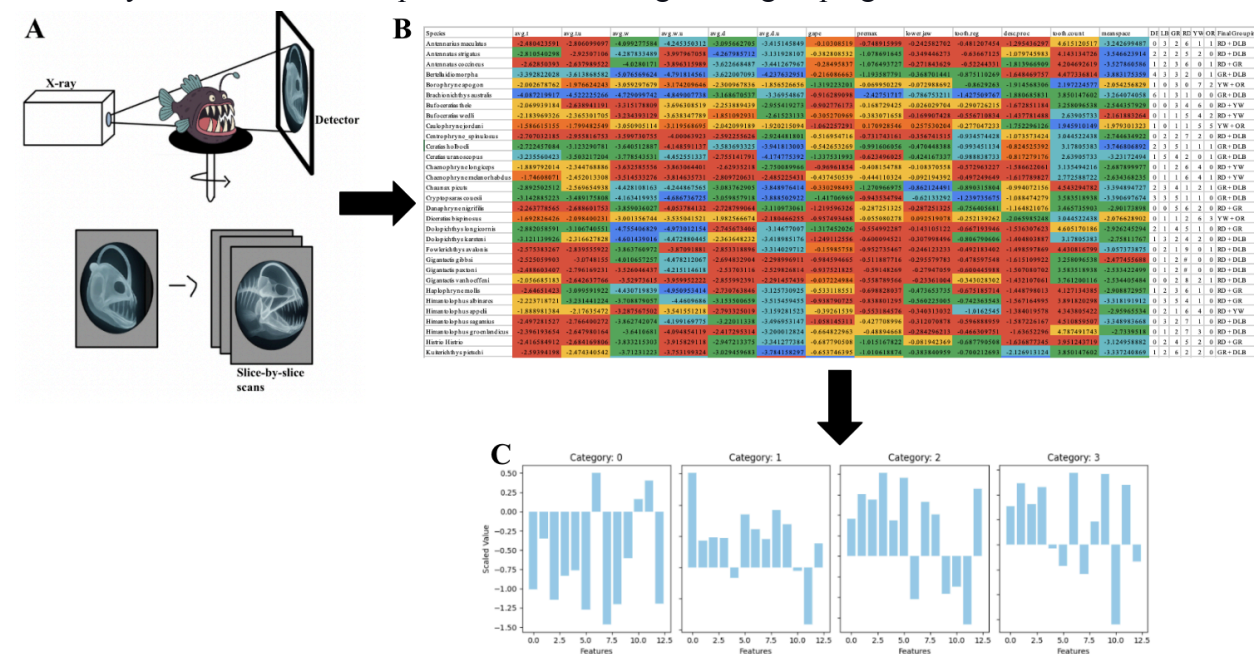
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### Lophiiformes Neural Network

AI and organisms are complex, emerging, and changing fields. AI enables researchers to understand animal behavior better (Urbaniak et al., 2024). Setting up different machine learning programs can allow scientists to explore and analyze behavioral traits that were previously inaccessible due to human interference. Another area where AI has aided biology is neural networks, which can learn, detect patterns, and accurately predict (Norouzzadeh et al., 2018). Typically, the more data the network can learn from, the more accurate it will be. Neural networks have been used to predict and identify different animals (BIOCLIP) by detecting the distinct patterns (Stevens et al., 2024). Animals can be very distinct and also appear very similar. One order of organisms that appears vastly different is the order Lophiiformes. This order comprises five suborders, four of which are benthic: *Antennarioidei*, *Chaunacoidei*, *Lophioidei*, and *Ogcocephaloidei*; the other order, the bathypelagic *Ceratioidei* (Hart et al., 2022), lives in the deep sea. All of these classes live in very different environments; however, many of the species live in the deep sea (Monterey Bay Aquarium, n.d.). Due to the scarcity of prey, deep-sea fishes are assumed to be opportunistic feeders with generalized diets (Ebeling & Cailliet, 1974). Recent comparative studies have shown that deep-sea fishes exhibit greater morphological diversity than other fishes (e.g., slope or shelf fishes) (Miller et al., 2024). These studies have used jaw morphotypes to examine the diets of shallow-marine fish, demonstrating that, in a high-productivity environment, these fish exhibit a wide variety of morphologies (Binning et al., 2009). Jaw and tooth morphologies are crucial for understanding the ecological functions and the operation of feeding traits in specific deep-sea habitats. There will be a distinct pattern in head and jaw measurements across families, enabling the neural network to predict family membership accurately.

The anglerfish data used in this study were obtained from an online repository (Dyrad), specifically from the research paper “*Many ways to build an angler: diversity of feeding morphologies in a deep-sea evolutionary radiation*” by Heiple et al. 2023. These researchers used 57 museum specimen anglerfish and subjected them to microCT (Figure 1A). To measure the anglerfish, they are placed in this machine, and slice-by-slice images are taken to measure various variables. The researchers transformed the raw data to account for 3D features, since the CT scanner only measures in 2D. Once downloaded, there were 22 measurements; after accounting for time, the measurements were reduced to 13. The thirteen measurements are: avg.t, avg. t.u, avg.w, avg. w.u, avg.d, avg. d.u, gape, premax, lower.jaw, tooth.reg, desc.proc, tooth.count, meanspace. Two different groupings with varying numbers of categories were created to compare the neural network’s accuracy. The data were first grouped by creating histograms for each measurement and marking which species were in each histogram bin (Figure 1B). The color-coded graph yielded five groups at the end, using all the specimens (Figure 1C). The other

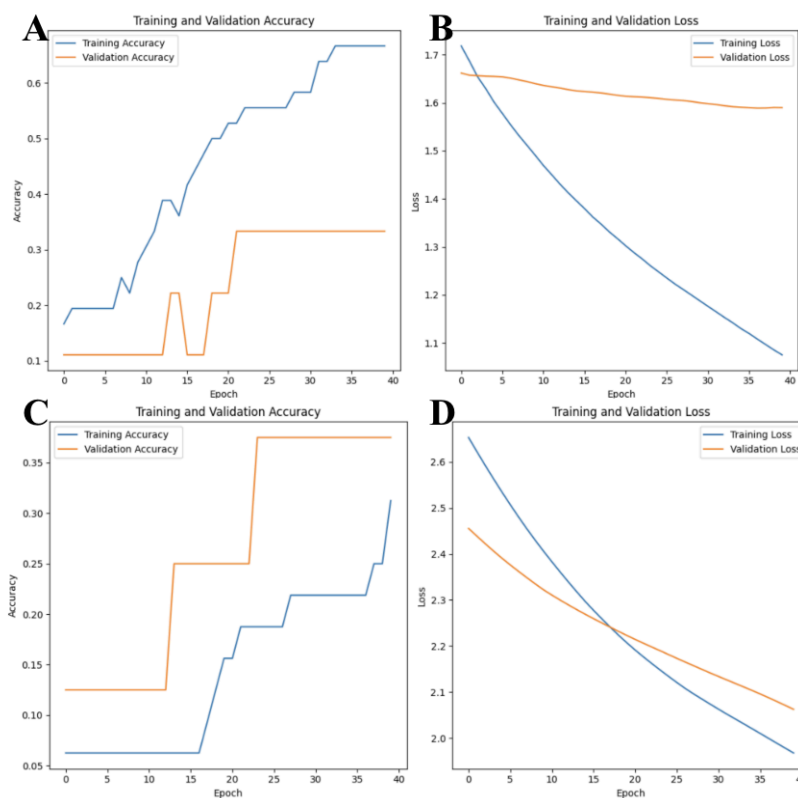
grouping was created by combining species according to the groupings in the supplementary data from Heiple et al.'s study. This grouping yielded 9 categories and used only 50 individuals rather than 57. The data was split into an 80/20 rather than the recommended 90/10 split because of the limited sample size. 80% of the specific species measurements were used for training, and the remaining 20% for testing. A confusion matrix will be created for both groups to determine how accurately the neural network predicts different anglerfish groupings.



**Figure 1.** Methods for processing anglerfish and data. A) microCT scanner is used to create slice-by-slice scans to analyze and measure varying bones, and ligaments. B) The transformed measurement data being organized into different groups for neural network training. C) Displaying some of the many different categories, bar graphs are displaying the different specimens within each category.

Before creating the confusion matrix, another figure can help illustrate the neural network's performance on the test data and its validation accuracy. Looking at the accuracy and loss of each group can help highlight whether the neural network is over- or under-fitting or a good fit. Training accuracy is the percentage of correct predictions on the training data, while validation accuracy measures the neural network's performance on a separate data set, the validation set. The training loss measures the difference between the network's predictions and the actual target values in the training data. In contrast, the validation loss measures how well the neural network performs on the validation data set that has not been used for training. The training and validation accuracies in the first group showed that the training accuracy reached ~90% (Figure 2A). The training validation, however, was not as high, ranging from 30% to 60%, and it ended at ~55%. The varying lines leave a large gap between them, leading to overfitting. The training loss for the first group decreases smoothly, indicating that the neural network is minimizing errors (Figure 2B). The validation loss, while decreasing, is less consistent than the

training loss and shows less improvement in error. In the second group, the training accuracy was only ~50% and showed a very stair-like pattern (Figure 2C). The training-validation appears very erratic, with large drops from ~38% to ~20%. These two lines are showing a pattern of underfitting. Finally, looking at the training loss from group two compared to the first group, it starts much higher at 2.2 rather than ~1.7; it also shows a similar pattern of a consistent decrease; however, it ends right after the first group starts at ~1.4 (Figure 2D). The validation loss converges earlier than the training loss, then decreases slightly; however, it decreases faster than the validation loss for the first group. Both of the groups showed interesting accuracy, with the lower number categories (5) having some overfitting and the higher number of categories (9) having underfitting.

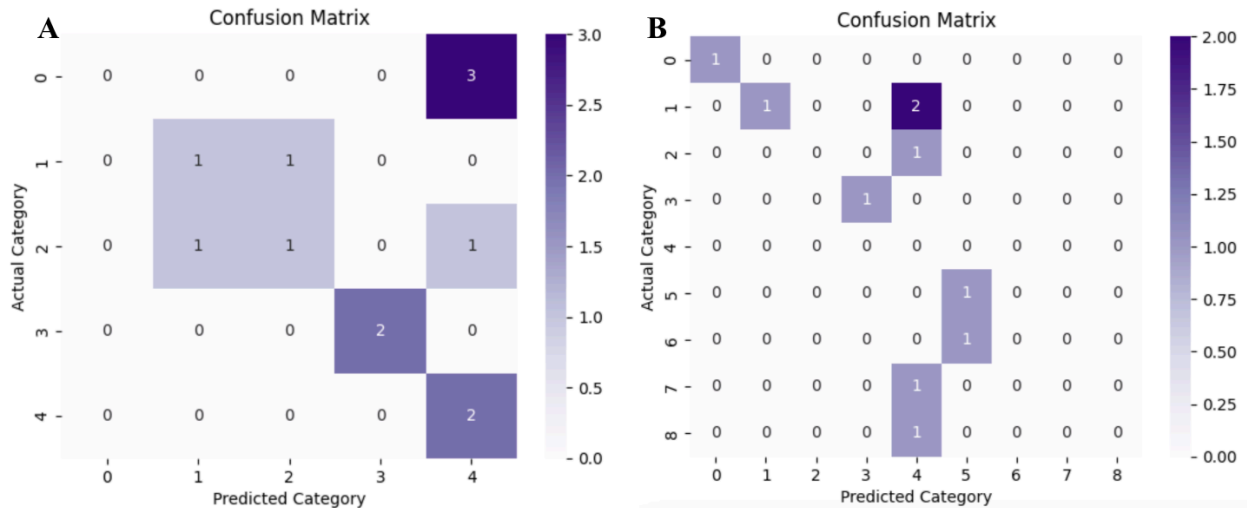


**Figure 2.** Training and validation accuracy and loss for both groups. A) 5 different groups showing a training accuracy of ~70% and validation accuracy of ~35%. B) 5 different groups showing the training loss starting at ~1.7 and ending around ~1.3, and the validation loss starting at ~1.7 and ending around ~1.6. C) 9 different groups showing the training accuracy ending at ~30% and the validation accuracy ending at ~37%. D) 8 different groups showing the training loss starting at ~2.65 and ending at ~2.0 and the validation loss starting at

~2.5 and ending at ~2.3.

A confusion matrix is an easy-to-read representation of the accuracy for each specific group, showing how well the neural network performed within each group. A perfect confusion matrix shows the highest value along the diagonal from the top left to the bottom right; this indicates a neural network that accurately predicted every test specimen. The first group's confusion matrix shows that the neural network performed very well on group 0 (Figure 3A). Looking at groups 1, 2, 3, and 4, the network did not do too well, mispredicting twice in group 2

and not predicting groups 1, 3, and 4 as accurately, since they have a value of 2, while the highest is 3. The second group, tasked with predicting 9 categories, accurately predicted only group 1 (Figure 3B). All of the other categories were not predicted correctly.



**Figure 3.** Confusion matrix of both groups. A) With 5 different categories the network accurately predicted group 0 with a value of 3, some mistakes in groups 3 and 4 with a value of 2, groups 1, 2, and 3 were not accurately predicted with a value of 1. B) With 8 different categories the network accurately predicted group 1 with a value of 2, 50% accurate on groups 0, 3, and 5 with a value of 1, and was not able to predict groups 2, 4, 6, 7, and 8.

Overall, the neural networks were unable to predict the different anglerfish groups accurately. The first group, organized by histogram generalizations, showed that the neural network was overfitting and only learned the measurements of the given species rather than the overall patterns. This means the data groups were too complex, and the network learned the noise, resulting in high accuracy on the training data but lower accuracy on new data (Sahoo, 2024). In the other group, the model was underfitting, meaning the model was too simple and unable to learn general patterns, resulting in poor performance on both the training and test data. Although group 1 showed overfitting, the confusion matrix indicated that it still performed well on new data. It accurately predicted group 0 and somewhat accurately predicted groups 1, 3, and 4, but quite poorly on group 2. At the same time, the other groups' confusion matrix showed an underfitting model: the network really only understood the pattern for group 1 and tried to predict almost every other group as group 1. In the future, this experiment should be redone by reprocessing the groups. From this information, knowing that the first group was too complex and the second group was too simple, finding an intermediate group will provide the network with the best possible data to detect a pattern without over- or under-fitting. Another factor that will improve the model's accuracy is increasing the sample size. It is difficult to collect lophiiform samples; however, the greater the number, the more likely the model will be accurate. If this neural network was to work it would greatly improve the field of marine biology, helping with potentially introducing new or unknown species or figuring out other patterns that maybe weren't visible before.

AI Statement: No AI was used to create this paper, grammarly was used to help with punctuation and grammar.

## Resources:

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